ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C7 · MAGNETS

Attract or not?

★☆☆
Grade 4

Learning goals

- Predict which materials are magnetic.
- Use pole rules.

1 Sort the objects. [8]

SORT THESE

plastic ruler **aluminium can** **iron nail** **steel paper clip** **wooden pencil**
steel spoon **gold ring** **copper wire**

Attracted to a magnet	Not attracted
steel paper clip	aluminium can
iron nail	copper wire
steel spoon	gold ring
	plastic ruler
	wooden pencil

2 Attract or repel? [4]

Poles facing each other	Attract or repel?
N and S	attract
N and N	repel
S and S	repel
S and N	attract

3 True or false? [4]

Statement	True	False	If false, correct it
1. All metals are magnetic.		✔	Only some metals, such as iron, steel and nickel, are magnetic.
2. A magnet can attract a paper clip through paper.	✔		
3. An electromagnet stops working when the electricity is switched off.	✔		
4. A compass needle points north.	✔		

4 Fill in the gaps. [4]

WORD BANK

attract plastic compass wire poles repel

The ends of a magnet are called poles. Like poles repel. A compass helps us find north. An electromagnet is made with a coil of wire around an iron nail.

ANSWER KEY

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THINK IT THROUGH · C7 · MAGNETS

Magnet puzzles

★★☆
Grade 4

Learning goals

- Explain magnet behaviour.
- Use electromagnets to solve problems.

1 **Rafi has a bag of mixed aluminium and steel cans for recycling. How could he separate them quickly? Explain.** [2]

Use a magnet: it attracts the steel cans but not the aluminium ones.

2 **A magnet has lost its N and S labels. How could you find which end is north, using a labelled magnet?** [2]

Bring the N pole of the labelled magnet near one end. If it repels, that end is N; if it attracts, that end is S.

3 **Why does a scrap-yard crane use an electromagnet and not an ordinary magnet?** [2]

It can be switched off to drop the scrap where it is needed; an ordinary magnet would not let go.

4 **Suggest two ways to make an electromagnet stronger.** [2]

Wrap more coils of wire; use more cells (more electricity); use a thicker iron core.

5 **Draw a magnet picking up a chain of paper clips. Label the poles and show where the force is strongest.** [2]

Magnet and clips

Bar magnet with N and S at the ends; clips hanging from the poles, not the middle.

ANSWER KEY

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INVESTIGATE · C7 · MAGNETS

How strong are our magnets?

★★☆
Grade 4

Learning goals

- Plan a fair test.
- Record and present results.
- Draw conclusions.

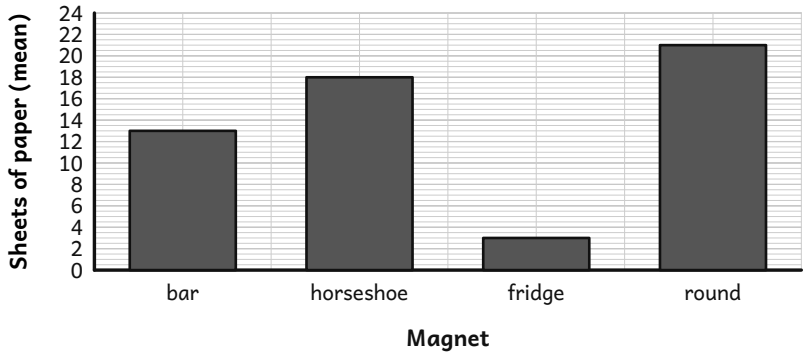
Class 4B tested four magnets. They counted how many sheets of paper each magnet could hold a steel paper clip through.

1 **Complete the plan.** [3]

Change	the magnet
Measure	number of sheets of paper the force passes through
Keep the same	the same paper clip, same paper, same way of holding the magnet

2 **Complete the table and draw a bar chart.** [4]

Magnet	Try 1	Try 2	Try 3	Mean
bar magnet	12	14	13	13
horseshoe magnet	18	17	19	18
fridge magnet	3	3	3	3
small round magnet	20	22	21	21



3 **Which magnet was strongest? Was it the biggest?** [2]

The small round magnet (21 sheets). No: it was small, so size does not tell you the strength.

4 **Why did they do three tries for each magnet?** [1]

To make the results more reliable and find a mean.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C7 · MAGNETS

Giant magnets in the ship-breaking yards

★★★
Grade 4

Learning goals

- See how magnets are used in industry.
- Use numbers.
- Think about workers' safety and recycling.

Science · Economics · Environment · Ethics

Along the beach at **Sitakunda**, near Chattogram, old ships from all over the world are taken apart in some of the biggest **ship-breaking yards** in the world. The steel from the ships is cut into pieces and **recycled**. Much of the steel used for building in Bangladesh comes from these old ships.

In steel yards and scrap yards, cranes with huge **electromagnets** lift piles of iron and steel. When the operator switches the current on, the scrap jumps up to the magnet; switching it off drops the scrap onto a truck. Ship breaking gives many people jobs, but it can be **dangerous** for workers and can **pollute** the beach and sea with oil and chemicals, so there are rules to make it safer and cleaner.

Based on reports about the ship-recycling industry in Bangladesh.

1 **Why is an electromagnet useful for lifting scrap steel?** [2]

It attracts the magnetic steel and can be switched off to drop it exactly where it is needed.

2 **Would the electromagnet pick up the copper cables and aluminium parts from a ship? Why?** [2]

No, because copper and aluminium are not magnetic.

3 **Maths. A crane magnet lifts 2 tonnes of steel each time. How many lifts are needed to load 30 tonnes?** [1]

30 ÷ 2 = 15 lifts

4 **Recycling ship steel is good, but there are problems. Complete the table.** [2]

Good points	Problems
saves iron ore and energy; cheap steel for buildings; jobs	dangerous work; oil and chemicals pollute the beach and sea

5 **Suggest two rules that would make ship breaking safer for workers.** [2]

e.g. helmets, gloves and boots for everyone; training; no children working; remove oil and gas safely before cutting.

ANSWER KEY

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REVIEW AND CHALLENGE · C7 · MAGNETS

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

L	A	O	U	O	F	U	F	Y	K
C	B	T	A	M	U	K	Y	Y	F
O	U	K	T	S	A	C	A	W	R
M	A	I	D	R	T	G	B	I	O
P	P	O	L	E	A	E	N	G	D
A	I	R	P	D	C	C	E	E	I
S	S	O	U	T	H	D	T	L	T
S	I	H	M	R	E	P	E	L	P
D	K	W	O	L	I	R	O	N	W
F	E	N	O	R	T	H	P	G	N

attract compass iron magnet north pole repel south steel

2 Which object would a magnet attract? [1]

- ○ a copper coin
- ○ an aluminium can
- ● **an iron nail**
- ○ a glass marble

3 Two north poles are pushed together. They will... [1]

- ○ attract
- ● **repel**
- ○ stick together
- ○ become weaker

4 Design a game that uses magnets (e.g. a fishing game). Draw it and explain how the magnets work. [3]

My magnet game

e.g. paper fish with steel paper clips; a rod with a magnet on a string attracts them.

5 **Challenge. Explain how you could make an electromagnet that is strong enough to lift 10 paper clips, and how you would test it fairly.** [2]

Wrap many coils of wire around an iron nail and use two cells. Test by counting how many clips it lifts with 10, 20 and 30 coils, keeping the same nail, clips and cells.